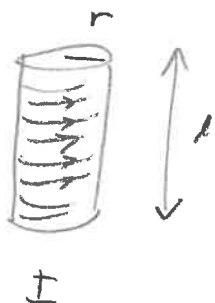


①.



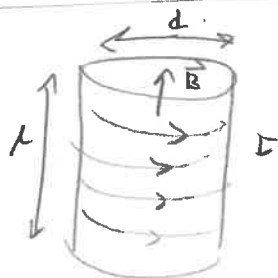
solenoid:  $N = 440$  turns,  $l = 25 \text{ cm}$   
 $r = 1.7 \text{ cm}$   
 $A = \pi r^2$   
 $n = \frac{N}{l}$

a)  $L = \mu_0 n^2 A l$   
 $= \mu_0 \frac{N^2 A}{l}$   
 $= \boxed{0.88 \text{ mH}}$

b)  $\mathcal{E} = -L \frac{di}{dt}$  solve for  $\frac{di}{dt} \rightarrow \frac{di}{dt} = -\frac{\mathcal{E}}{L}$

For  $\mathcal{E} = 70 \text{ mV}$ ,  $\left| \frac{di}{dt} \right| = \frac{\mathcal{E}}{L} = \boxed{79.2 \frac{\text{A}}{\text{s}}}$

②.



$I = 36.5 \text{ mA}$   
 $N = 400$  turns  
 $d = 16.5 \text{ mm}$   
 $l = 11.5 \text{ cm}$

a)  ~~$B = \mu_0 I n$  (constant inside solenoid)~~

$\rightarrow n = \frac{N}{l}, A = \pi \left( \frac{d}{2} \right)^2$

a)  $B = \mu_0 I n$  (constant inside solenoid, pointing up ward)  
 $= 1.6 \times 10^{-4} \text{ T} = \boxed{160 \mu\text{T}}$

b)  $\Phi_B = N B A = \boxed{13.6 \times 10^{-6} \text{ T} \cdot \text{m}^2}$

For single turn  $\Phi_B \Big|_{N=1} = B A = \boxed{3.41 \times 10^{-8} \text{ T} \cdot \text{m}^2}$

c) Inductance:

$L = \mu_0 n^2 A l = 3.7 \times 10^{-4} \text{ H} = \boxed{0.37 \text{ mH}}$

d) magnetic field and magnetic Flux depend on current

(2)

(3)  $L = 70 \text{ mH}$  inductor  
 $i(t) = 3t^2 - 7t$  where  $t$  is in seconds

a)  $\mathcal{E} = -L \frac{di}{dt} = -L(6t - 7)$

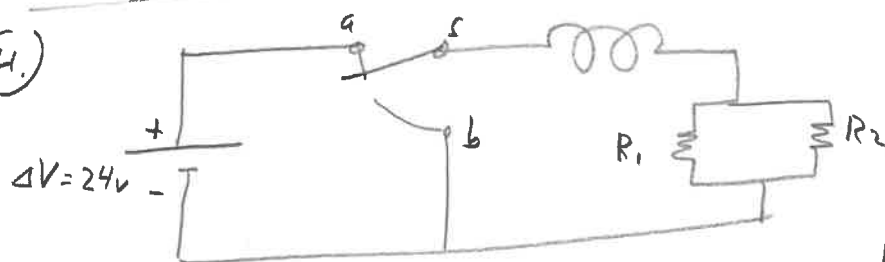
Magnitude:  $|\mathcal{E}(t)| = L|(6t - 7)|$

At  $t = 1 \text{ sec}$ ,  $|\mathcal{E}(t=1 \text{ sec})| = L \times \frac{1 \text{ Amp}}{s} = \boxed{70 \text{ mV}}$

b) At  $t = 4 \text{ sec}$ ,  $|\mathcal{E}(t=4 \text{ sec})| = L \times \frac{17 \text{ A}}{s} = \boxed{1,190 \text{ mV}}$

c)  $\mathcal{E} = 0$  when  $6t - 7 = 0 \rightarrow \boxed{t = \frac{7}{6} \text{ sec}}$

(4)



$$L = 4.65 \text{ mH}$$

$$R_2 = 445 \Omega$$

$$\tau_L = 14.9 \mu\text{s}$$

a)  $\tau_L = \frac{L}{R_{\text{eff}}}$  where  $R_{\text{eff}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$

$$\rightarrow R_{\text{eff}} = \frac{L}{\tau_L} = \boxed{312 \Omega} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_{\text{eff}}} \rightarrow \frac{1}{R_1} = \frac{1}{R_{\text{eff}}} - \frac{1}{R_2}$$

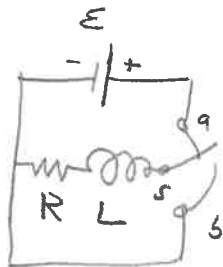
$$= \frac{R_2 - R_{\text{eff}}}{R_2 R_{\text{eff}}}$$

So  $R_1 = \frac{R_2 \cdot R_{\text{eff}}}{R_2 - R_{\text{eff}}} = \boxed{1,044 \text{ k}\Omega}$

b) Current is maximum when switch throws to b

$$I = \frac{\Delta V}{R_{\text{eff}}} = \boxed{77 \text{ mA}}$$

(5.)



$$L = 140 \text{ mH}$$

$$R = 5.2 \Omega$$

$$\varepsilon = 6.0 \text{ V}$$

(3)

a) After switch is thrown to a

$$i(t) = \frac{\varepsilon}{R} (1 - e^{-t/\tau_L}) \quad , \quad \tau_L = \frac{L}{R} = \boxed{27 \text{ ms}}$$

Find  $t$  such that  $i(t) = 220 \text{ mA}$ :

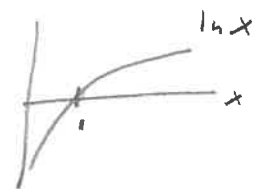
$$\frac{R i(t)}{\varepsilon} = 1 - e^{-t/\tau_L}$$

$$\rightarrow e^{-t/\tau_L} = 1 - \frac{R i(t)}{\varepsilon}$$

$$-t/\tau_L = \ln \left( \frac{\varepsilon - R i(t)}{\varepsilon} \right)$$

$$t = -\tau_L \ln \left( \frac{\varepsilon - R i(t)}{\varepsilon} \right)$$

$$= \tau_L \ln \left( \frac{\varepsilon}{\varepsilon - R i(t)} \right) = \boxed{5.7 \text{ ms}}$$



$$b) \quad i(t=10 \text{ s}) = \frac{\varepsilon}{R} (1 - e^{-10 \text{ s}/\tau_L})$$

$$= \boxed{1.15 \text{ A}}$$

c) Switch thrown from a to b:

$$i(t) = \frac{\varepsilon}{R} e^{-t/\tau_L}$$

$$\text{Want } i(t) = 160 \text{ mA}$$

Solve for  $t$ :

$$e^{-t/\tau_L} = \frac{R i(t)}{\varepsilon}$$

$$-t/\tau_L = \ln \left( \frac{R i(t)}{\varepsilon} \right)$$

$$\rightarrow t = \tau_L \ln \left( \frac{\varepsilon}{R i(t)} \right) = \boxed{53 \text{ ms}}$$

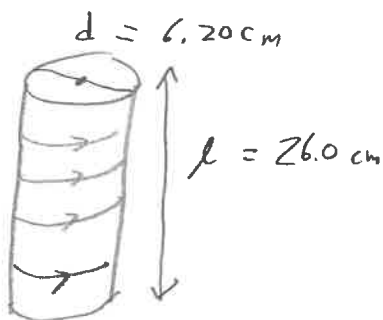
(4)

(6)  $E = 147 \frac{V}{m}$  ,  $B = 4,5 \times 10^{-5} T$

a)  $u_E = \frac{1}{2} \epsilon_0 E^2 = \boxed{95,6 \text{ nJ/m}^3}$

b)  $u_B = \frac{1}{2\mu_0} B^2 = \boxed{806 \mu\text{J/m}^3}$

(7)  $B = 4,9 T = \text{const}$



a)  $u_B = \frac{1}{2\mu_0} B^2 = \boxed{9,55 \times 10^6 \frac{J}{m^3}}$

b)  $U_B = u_B \cdot \text{Volume}$   
 $= u_B \pi \left(\frac{d}{2}\right)^2 \cdot l$   
 $= \boxed{7,500 J} = \boxed{7,5 kJ}$

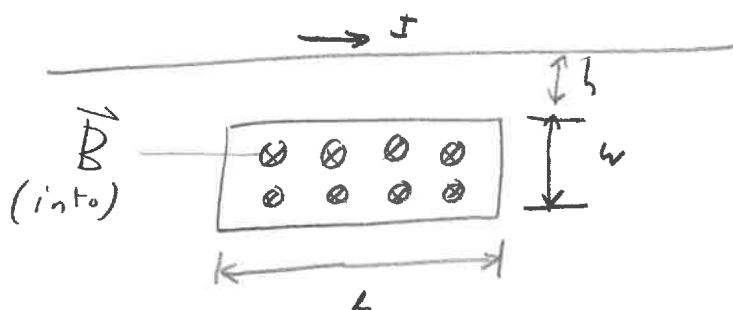
(8)  $M = 130 \mu H$

$i(t) = 15 \sin(1,25 \times 10^3 t) = i_0 \sin(\omega t)$   $i_0 = 15 A$   $\omega = 1,25 \times 10^3 \frac{rad}{s}$

$\mathcal{E} = -M \frac{di}{dt} = -M i_0 \omega \cos(\omega t)$

$\mathcal{E}_{peak} = M i_0 \omega = \boxed{2,44 V}$

(9)



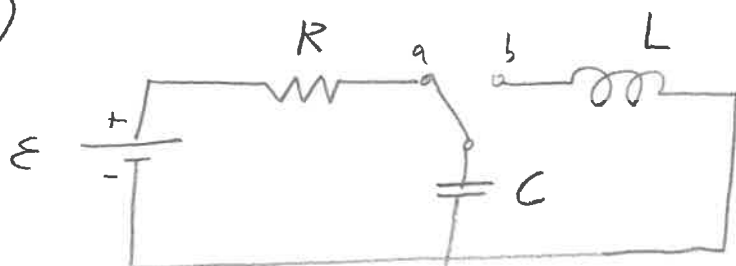
$h = 0,4 \text{ mm}$   
 $w = 1,1 \text{ mm}$   
 $l = 2,7 \text{ mm}$

$M = \frac{\Phi_B}{I}$

$\Phi_B = \int \vec{B} \cdot \vec{n} da = \int_{r=h}^{h+w} dr l \frac{\mu_0 I}{2\pi r}$   
 $= \frac{\mu_0 I l}{2\pi} \ln\left(\frac{h+w}{h}\right)$

$\rightarrow M = \frac{\mu_0 l}{2\pi} \ln\left(\frac{h+w}{h}\right) = \boxed{714 \text{ pH}}$  where  $\mu H = 10^{-12} H$

(10.)



$$\begin{aligned}
 R &= 10 \Omega \\
 L &= 0.1 \text{ H} \\
 C &= 1.1 \mu\text{F} \\
 \mathcal{E} &= 12 \text{ V}
 \end{aligned}$$

(5)

a)  $\omega = \frac{1}{\sqrt{LC}}$  ,  $F = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} = \boxed{480 \text{ Hz}}$

b) max charge on capacitor  
 $\Delta V = \mathcal{E}$  ,  $Q_{\text{max}} = C \Delta V = C \mathcal{E} = \boxed{13.2 \mu\text{C}}$

c) Energy transferred back and forth between capacitor and inductor

$$E = \frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} L I^2$$

where  $E = \frac{1}{2} \frac{Q_{\text{max}}^2}{C} = \frac{1}{2} C \mathcal{E}^2 = \text{const}$

Get maximum current when  $Q = 0$

$$\rightarrow \frac{1}{2} \frac{Q_{\text{max}}^2}{C} = \frac{1}{2} L I_{\text{max}}^2$$

$$I_{\text{max}} = \frac{Q_{\text{max}}}{\sqrt{LC}} = \boxed{\omega Q_{\text{max}}} = \boxed{40 \text{ mA}}$$

d) total energy = const (so has same value at all times)

$$E = \frac{1}{2} C \mathcal{E}^2 = \boxed{79.2 \mu\text{J}}$$

(11.)

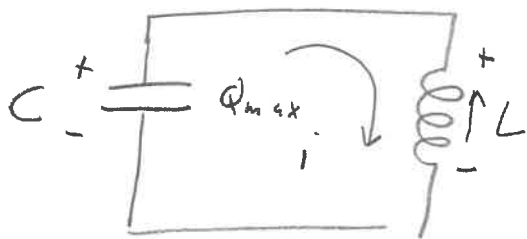
$$L = 3.3 \text{ H}$$

$$C = 838 \text{ pF}$$

$$Q_{\max} = 128 \text{ } \mu\text{C}$$

$$t = 4.00 \text{ ms}$$

$$\omega = \frac{1}{\sqrt{LC}}$$



$$i = -\frac{dQ}{dt}$$

$$\mathcal{E}_L = -L \frac{di}{dt}$$

$$Q = Q_{\max} \cos(\omega t), \quad \omega = \frac{1}{\sqrt{LC}}$$

$$i = -\frac{dQ}{dt} = Q_{\max} \omega \sin(\omega t) = i_{\max} \sin(\omega t)$$

$$\text{where } i_{\max} = Q_{\max} \omega$$

$$a) \quad U_C = \frac{1}{2} C \Delta V^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$U_C(t) = \frac{1}{2} \frac{Q_{\max}^2}{C} \cos^2(\omega t) = \boxed{6.04 \text{ J}}$$

$$b) \quad E = \frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} L I^2 = \text{const}$$

$$E = \frac{1}{2} \frac{Q_{\max}^2}{C} = \boxed{9.78 \text{ J}}$$

$$c) \quad U_L(t) = \frac{1}{2} L I(t)^2 = E - \frac{1}{2} \frac{Q^2(t)}{C}$$

$$= E - U_C(t)$$

$$= \boxed{3.73 \text{ J}}$$

(using answers from (a) and (b))